

CHENNA SAISANDEEP

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PROFILE

Machine Learning Engineer focused on building robust AI systems, agentic workflows, and domain-specific language models. Experienced in designing end-to-end ML pipelines, optimizing inference performance, and deploying intelligent applications. Passionate about solving complex engineering challenges through applied AI.

EDUCATION

Chennai Mathematical Institute <i>Master's degree in Data Science</i> Current GPA: 9.22	Aug 2023 – Present
Indian Statistical Institute <i>Bachelor's degree in Mathematics (Hons.)</i> Aggregate Percentage: 77% (1st Division with Hons.)	Jul 2019 – May 2022

EXPERIENCE

Shree Cement Ltd. (Kolkata) <i>Data Scientist Intern</i> - Improved solar power generation efficiency by designing a near real-time data-driven anomaly detection system. - Applied machine learning models including Isolation Forest, Local Outlier Factor, DBSCAN, One-Class SVM, and neural networks. - Integrated the application seamlessly with Power BI to enable visual inspection and operational analysis.	May 2024 – Jul 2024
Cloudcraftz AI (Kolkata) <i>Data Scientist Intern</i> - Evaluated large-scale tabular synthetic data generated by various GAN models using robust statistical metrics. - Leveraged explainable AI (XAI) to interpret complex models such as deep neural networks and ensemble models. - Optimized the BodhiX web application by implementing multiprocessing, significantly reducing computation times.	May 2023 – Jul 2023

PROJECTS

AgentMesh: Agentic Workflow Orchestrator <i>TypeScript, Python, Fastify, React Flow, PostgreSQL, Redis, LangGraph</i> - Built a graph-native platform for composing, executing, and observing AI-enabled workflows with real-time control loops. - Architected a hybrid system using TypeScript for API/contracts and Python for DAG execution and validation. - Implemented real-time execution telemetry via Redis pub/sub and node-level caching to avoid redundant computation.
GraphRAG Planning Platform <i>Python, FastAPI, TypeScript, Neo4j, Qdrant, LangChain</i> - Developed an AI-assisted planning platform using a hybrid retrieval strategy (semantic and sparse search) fused with reciprocal rank fusion. - Integrated neural reranking and structural context from a Neo4j graph database to map feature requests to implementation plans. - Delivered a REST API and a VS Code extension with streaming progress updates and durable Redis-backed queue execution.
Domain-Specific SLM Training Pipeline <i>Python, PyTorch, Transformers, TRL, SFT, DPO</i> - Engineered an end-to-end data and alignment pipeline to specialize a compact language model for embedded firmware domains. - Implemented staged model adaptation (continual pre-training, supervised fine-tuning, and direct preference optimization) handling long contexts. - Created callback-based evaluation tools to compare training checkpoints interactively.

SKILLS

Languages & DBs	Python, TypeScript, SQL, PostgreSQL, Neo4j, Qdrant, Redis
Machine Learning	TensorFlow, PyTorch, Scikit-learn, LangGraph, Transformers, LLM Fine-Tuning, RAG
Frameworks & Tools	FastAPI, Fastify, React, Docker, Git, VS Code Extensions
Data Science	Pandas, NumPy, Matplotlib, OpenCV, Statistical Modeling, Outlier Detection

ACHIEVEMENTS & CERTIFICATIONS

- MSQMS (ISI) 2023:** All India Rank 4. Awarded Indian Statistical Institute Student Fellowship.
- BMATH (ISI) 2019:** All India Rank 18. Awarded Indian Statistical Institute Student Fellowship.
- Certifications:** Google Data Analytics Professional Certificate, Complete Data Science Bootcamp (Udemy).